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Keysight Technologies  
South Queensferry, Edinburgh, United Kingdom

Dear Hiring Manager,

I am writing to apply for the internship on O-RAN digital twins and AI/ML-driven RAN optimisation. My MSc by Research at the University of Edinburgh, supervised by Prof. Mahesh K. Marina, sits on precisely the two halves that this role combines. I build the digital twin and emulation platforms on which RAN algorithms are developed and validated, and I design the near-real-time policies that decide how those platforms use shared compute. In my work these have never been two separate tasks, which is why a role that asks for both is the one I want.

My thesis project, Flux-DT, is an elastic digital twin runtime for AI-RAN. Instead of observing a network from the outside, the twin runs inside the radio path of a live OpenAirInterface 5G stack, so that real signalling and user traffic flow through a ray-traced channel model computed across GPU workers. On top of it I designed a control loop that reacts to radio-driven resource pressure on the timescale the radio imposes: it relocates work to less contended machines, lowers simulation fidelity only within bounds that live radio procedures tolerate, and suspends and resumes the twin when no further reduction is safe. Getting this right meant treating the live RAN as the priority workload at every step, and treating the twin's own state as something that must survive being interrupted. I understand these to be the same constraints that a production digital twin platform has to respect.

I have also validated systems of this kind at scale. On EmuRAN, a cloud-based RAN emulation system conditionally accepted at ACM MobiCom 2026, I owned the evaluation: I deployed and operated the emulator on public cloud at 130 instances and 6,400 CPU cores, emulating 10,000 mobile devices and 10,000 base stations, and I designed the fidelity and scalability experiments that established it behaves like a real RAN rather than merely running at scale. My machine learning systems work is closely connected to this. Morphling, an emulator for evaluating distributed machine learning training on heterogeneous edge devices, received the Best Paper Award at EdgeSys 2026, and on Weaver I evaluated a controller that decides how much accelerator capacity an AI workload may harvest from shared RAN infrastructure without degrading radio performance. Alongside this, my C and C++ experience comes from compiler research, where I built an LLVM-based auto-quantization framework and evaluated it on embedded ARM devices.

What draws me to Keysight specifically is that this role puts algorithm development and platform work in the same place. Developing AI and machine learning algorithms for prioritised O-RAN use cases, and then validating them on digital twins and emulation platforms instead of in a conventional lab, is exactly the loop I have been building the tools for, and I would much rather close that loop against real equipment than in simulation alone. Applying AI and machine learning to the optimisation of cellular network resources, and doing so in the context of 5G and future 6G radio access, is the direction I intend to build my career in.

I graduate in September 2026 and am available for a full-time internship from then. I am already based in Edinburgh, so I can work on site without relocating. I would welcome the opportunity to discuss how I could contribute to the team, and I am happy to talk through any of the systems above in more detail.

Yours faithfully,

**Yufeng (Alex) Xia**